

KUNWAR SHAURYA PRATAP SINGH

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Penultimate-year CS student specializing in distributed systems, concurrent infrastructure, and AI-integrated software. Seeking to build high-performance, large-scale systems at Google.

EDUCATION

Vellore Institute of Technology

Vellore, India

B.Tech in Computer Science and Engineering – AI & Data Engineering

Aug. 2024 – May 2028

- CGPA: **9.02/10**; penultimate-year student, eligible for the Summer 2027 internship cycle.
- Coursework: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, Machine Learning, Computer Architecture, Distributed Systems, Concurrency & Synchronization.

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, TypeScript, SQL, Solidity

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, Distributed Systems, Concurrency & Synchronization, Software Design, Synchronization Primitives

Systems & Backend: FastAPI, Node.js, Express.js, PostgreSQL, Redis, REST APIs, Socket.IO, Async I/O, Multi-threading, Web3.py

AI/ML: TensorFlow Lite Micro, XGBoost, SHAP, LangChain, FAISS, RAG Pipelines, Groq API, Scikit-learn, Pandas, NumPy

Tools & Platforms: Git, GitHub, Docker, GitHub Actions, Postman, Hardhat, Vercel, Railway

Frontend: React.js, Next.js, D3.js, Vite, Tailwind CSS

EXPERIENCE

MyPerro

Remote / VIT-TBI

Software Developer

Aug. 2024 – Present

- Built a **real-time IoT data ingestion pipeline** processing sensor telemetry from smart collar hardware into a responsive client layer, handling **continuous event streams** with sub-200ms end-to-end latency on the app surface.
- Reduced **unnecessary re-render cycles** by **~40%** through memoization and fine-grained state management, cutting perceived load time across mobile and web production flows; diagnosed bottlenecks using React DevTools profiler.

IEEE Computer Society – VIT

Vellore, India

Senior Core Member – Technical

Feb. 2025 – Present

- Architected and deployed **event infrastructure systems** serving **500+ concurrent users** across hackathons, including real-time registration pipelines with **zero downtime** during peak load spikes.
- Led backend integration for participant-facing platforms using **Node.js/Express with multi-threaded, queue-based job processing**; designed fault-tolerant API contracts and coordinated system reliability and live-event debugging across teams.

PROJECTS

SentinelMesh – Distributed Edge AI Cybersecurity Mesh | C++, Python, ESP32, TensorFlow Lite Micro, Node.js 2026

- Architected a **fault-tolerant, distributed mesh** across ESP32 nodes using **ESP-NOW** for concurrent peer-to-peer threat-intelligence broadcasts; mesh remains **fully operational with up to 40% node failure**, validated under simulated compromise scenarios.
- Implemented **multi-threaded, on-device concurrent inference** in C++ with TensorFlow Lite Micro, classifying packet anomalies locally without cloud dependency; achieved **sub-10ms classification latency** per node – **12× faster** than equivalent cloud round-trip (~120ms).
- Built a **high-throughput real-time monitoring pipeline** with Node.js and Socket.IO sustaining **>1,000 events/sec**; tuned event-loop throughput via Node.js `--prof` to eliminate I/O bottlenecks; anchored SHA-256 hashes to Polygon testnet via Web3.py for tamper-proof audit trails.

ExamChain – Zero-Trust Distributed Examination Infrastructure | FastAPI, PostgreSQL, Redis, Solidity, Web3.py 2026

- Designed a **scalable, distributed exam platform** with Merkle-root integrity locking and blockchain-backed result verification; **Redis-backed session caching** achieved **>95% cache hit rate**, reducing session lookup latency from ~80ms to <5ms under concurrent load.
- Implemented **async, high-concurrency backend** with FastAPI and async SQLAlchemy, supporting **200+ parallel simultaneous sessions**; profiled hotspots with cProfile, resolving a session-validation bottleneck that cut p99 API latency by 38%.
- Engineered on-chain Solidity smart contracts for **immutable result anchoring** and a React dashboard surfacing live integrity flags, chain status, and risk telemetry – end-to-end **zero-trust verification with no single point of failure**.

NetraAI – Distributed Mule Account Detection Platform | Python, XGBoost, SHAP, FastAPI, Streamlit 2026

- Built a **multi-layer fraud detection pipeline** using XGBoost with SHAP-based explainability and a Mule Lifecycle Classifier, achieving **91% F1-score** on held-out financial transaction data for real-time risk scoring.
- Architected an end-to-end **data pipeline and inference system** ingesting behavioral, transactional, and graph-based features at **>500 records/sec**; exposed predictions through a production-ready FastAPI service with **sub-50ms p99 latency**.

ACHIEVEMENTS

- **2nd Place, Alpha Forge 3.0 Hackathon** (*out of 200+ teams*): Built **Vivayu / AgriGPT** – an AI + IoT agricultural diagnostic system with multi-sensor data pipelines, XGBoost inference, LangChain + FAISS retrieval, and voice-first RAG workflows on Groq API.
- **Exemplary Performance, WomenTechies'26 – GDG On Campus VIT**: Engineered a full-stack technical solution under a **36-hour deadline** as part of **Team Togepi**, recognized for engineering depth and collaborative problem-solving.
- **Competitive Programming**: Solved **150+ DSA problems** in Java/C++ across arrays, trees, graphs, dynamic programming, binary search, greedy, and hashing; focused on medium-to-hard difficulty; actively practicing on LeetCode and Codeforces.